

$$\bar{x}^{(k)} = G \bar{x}^{(k-1)} + \bar{c}$$

$\bar{c}$ - given, $G_{n \times n}$ - given

$$\bar{x}^{(k)} \rightarrow (\bar{I} - G)^{-1} \bar{c} \quad \forall \bar{x}_0$$



$$\rho(G) < 1$$

spectral radius
of G is $\rho(G)$

$$\rho(G) = \lim_{n \rightarrow \infty} \|G^n\|^{1/n}$$

Corollary. The formula

$$Q \bar{x}^{(k)} = (Q - A) \bar{x}^{(k-1)} + \bar{b},$$

$$\bar{x}^{(k)} = Q^{-1} (Q - A) \bar{x}^{(k-1)} + \bar{b},$$

$$\bar{x}^{(k)} = (\bar{I} - Q^{-1}A) \bar{x}^{(k-1)} + \bar{b}$$

$$A\bar{x} = \bar{b}$$

$$0 = -A\bar{x} + \bar{b}$$

$$Q\bar{x} = Q\bar{x} - A\bar{x} + \bar{b}$$

$$Q\bar{x} = (Q - A)\bar{x} + \bar{b}$$

will produce a sequence converging
to the solution of $A\bar{x} = \bar{b}$
for any $\bar{x}^{(0)}$, $(\Rightarrow) \rho(I - D^{-1}A) < 1$.

Now look at Gauss-Seidel

(D - ~~upper~~ ^{lower} Δ part of A).

Thm If A is diagonally
dominant, then the G.-S. method
converges for any starting vector
Pf. Enough to show that $\rho(I - D^{-1}A) < 1$.

Let λ be an e-value of $I - D^{-1}A$,
and $\bar{x}$ be a corresponding e-vector.
Assume $\|\bar{x}\|_{\infty} = 1$.

$$(I - Q^{-1}A) \bar{x} = \lambda \bar{x}$$



$$Q\bar{x} - A\bar{x} = \lambda Q\bar{x}$$

Since Q is the lower Δ part of A ,

$$\begin{aligned} -\sum_{j=i+1}^n a_{ij} x_j &= \lambda \sum_{j=1}^i a_{ij} x_j \quad (1 \leq i \leq n) \\ &= \lambda a_{ii} + \lambda \sum_{j=1}^{i-1} a_{ij} x_j \end{aligned}$$

$$\Downarrow$$

$$\lambda a_{ii} = -\lambda \sum_{j=1}^{i-1} a_{ij} x_j - \sum_{j=i+1}^n a_{ij} x_j$$

Pick an index i such that $|x_i| = 1 \geq |x_j|$
 $j \neq i, 0 \leq j \leq n.$



$$|\lambda| |a_{ii}| \leq |\lambda| \sum_{j=1}^{i-1} |a_{ij}| + \sum_{j=i+1}^n |a_{ij}|$$

(167)

$$|\lambda| \leq \frac{\sum_{j=i+1}^n |a_{ij}|}{|a_{ii}| - \sum_{j=1}^{i-1} |a_{ij}|} \quad \checkmark \text{diag. dominance.} < 1$$

$$|a_{ii}| > \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}| = \sum_{j=1}^{i-1} |a_{ij}| + \sum_{j=i+1}^n |a_{ij}|$$

$$|a_{ii}| - \sum_{j=1}^{i-1} |a_{ij}| > \sum_{j=i+1}^n |a_{ij}| \Leftrightarrow 1 > \frac{\sum_{j=i+1}^n |a_{ij}|}{|a_{ii}| - \sum_{j=1}^{i-1} |a_{ij}|}$$

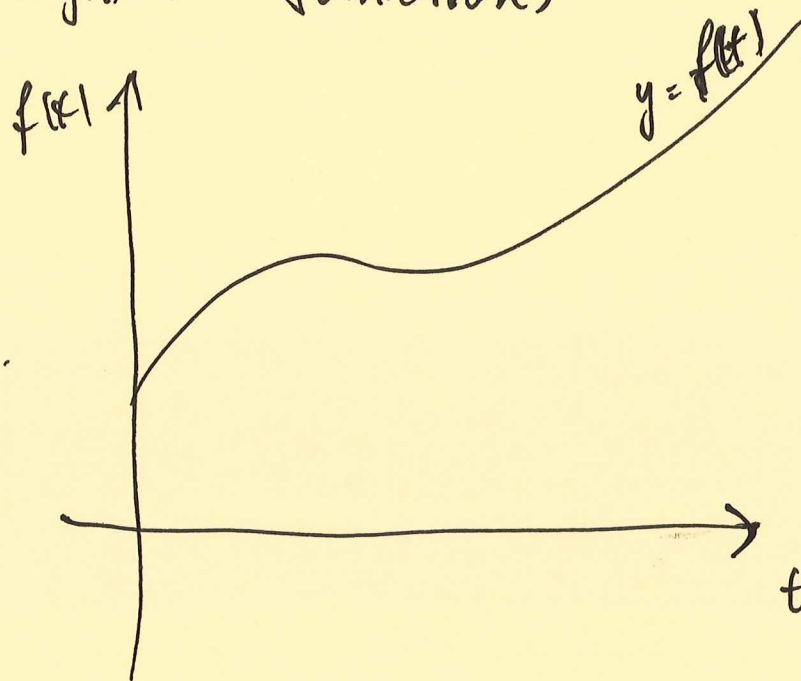
$$\rightarrow \max_i |\lambda_i| < 1 \quad (\text{finitely many } \lambda_i\text{'s})$$

$$\Downarrow$$

$$\rho(I - A^{-1}A) < 1.$$

□.

Analysis : functions



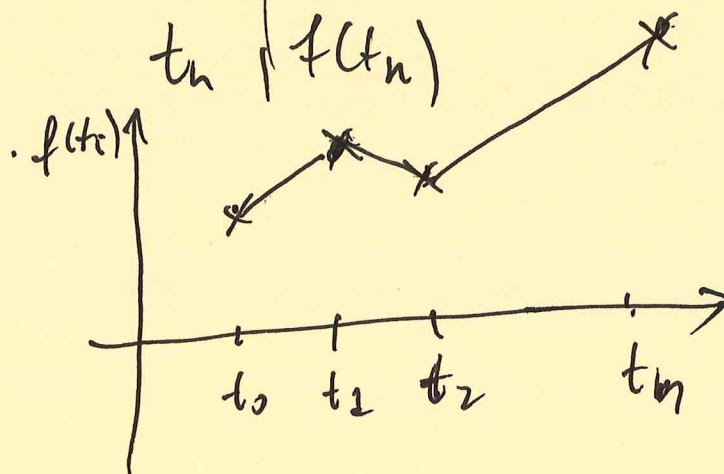
not possible.

OR

Numerics : finite table of values

t_i	$f(t_i)$
t_0	$f(t_0)$
$\vdots$	
t_n	$f(t_n)$

equation for f
may be
unavailable.



$$p(x) = \sum_{j=0}^n a_j x^j = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$$

a_j - real #s

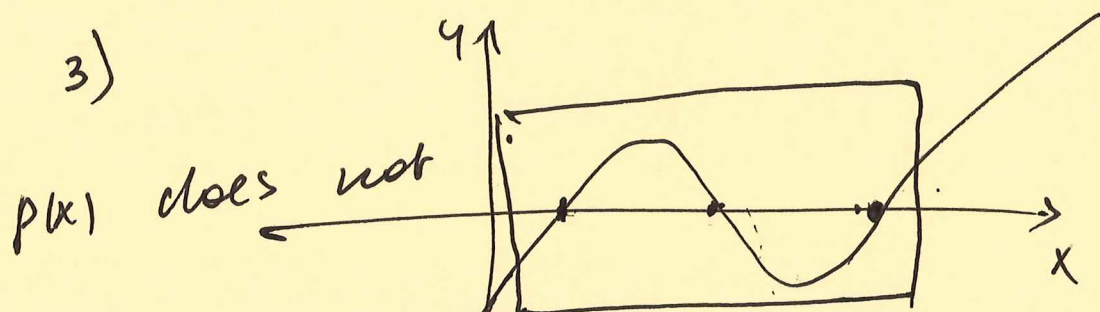
$j \geq 0$, $j \in \mathbb{Z}$.

$a_n \neq 0$

n -degree

1) $p(x)$ can have at most n real zeros

2) $p(x)$ is $C^\infty(\mathbb{R}) \Rightarrow$ continuous on $\mathbb{R}$



oscillate away from "zeros region"

4) tails are predictable
(degree, sign of a_n)

Interpolation.

Problem : given $x_0, x_1, \dots, x_n$
 $y_0, y_1, \dots, y_n$

find a polynomial P of the lowest possible degree for which

$$P(x_i) = y_i \quad i=0, 1, \dots, n$$

Such $p(x)$ is ^{called} the interpolant for the data.

Thm If $x_i, i=0, 1, \dots, n$ are distinct
 real #s, then $\forall$ ^{choice of} $y_0, y_1, \dots, y_n$

there is a unique polynomial
 P_n of degree at most n such that
 $P_n(x_i) = y_i, \quad 0 \leq i \leq n.$

Pf. 1. Uniqueness.

Suppose there were two such polynomials, p_n, q_n .

Then $(p_n - q_n)(x_i) = 0, 0 \leq i \leq n$.

Since $\deg(p_n - q_n) \leq n$, $p_n - q_n$ can have at most n zeros, unless $p_n - q_n \equiv 0$.
 Since # of x_i is $n+1$, $p_n = q_n$.

2. Existence. By induction

a) $n=0$, a constant polynomial p_0 can be chosen so that $p_0(x_0) = \overset{= p_0}{p_0(x_0)} = y_0$

b) Assume that there is p_{k-1} of degree $\leq k-1$ such that $p_{k-1}(x_i) = y_i$,
 $i=0, \dots, k-1$.

Look for p_k in the form

$$(*) \quad p_k(x) = p_{k-1}(x) + C \underbrace{(x - x_0)(x - x_1) \dots (x - x_{k-1})}_{C \text{ - adjustable.}}$$

$\deg p_k \leq k$

Note that

$$p_k(x_i) = y_i \quad i = 0, \dots, k-1$$

To ensure $p_k(x_k) = y_k$, need to choose C

$$p_k(x_k) = p_{k-1}(x_k) + C(x_k - x_0) \dots (x_k - x_{k-1}) = y_k$$

$$C = [y_k - p_{k-1}(x_k)] [(x_k - x_0) \dots (x_k - x_{k-1})]^{-1}.$$

(can be solved when $x_0, x_1, \dots, x_k$ are distinct)



Corollary: We can construct interpolation polynomial recursively

$$(x_0, y_0) \Rightarrow p_0(x) = y_0$$

$$(x_0, y_0), (x_1, y_1) \Rightarrow p_1(x) = p_0(x) + C_1(x - x_0)$$

$$\dots \Rightarrow p_2(x) = p_1(x) + C_2(x - x_0)(x - x_1)$$

$$p_{k+1}(x) = p_k(x) + C_k(x - x_0) \dots (x - x_{k-1}).$$

Lagrange interpolation.

Look for $P(x) =$

$$= \sum_{k=0}^n y_k l_k(x) \quad (**)$$

$l_k(x)$ polynomials that depend on $x_0, x_1, \dots, x_n$, but not on $y_0, y_1, \dots, y_n$

Want $(**)$ to work for any choice of $y_0, y_1, \dots, y_n$ (with the same $l_k(x)$)

Consider a special choice of

$y_0, y_1, \dots, y_n$: all 0 except $y_i = 1$

then $P(x_j) = \sum_{i=0}^n y_i l_i(x_j) = \begin{cases} 0 & \text{if } j \neq i \\ 1 & \text{if } j = i \end{cases}$

$j = 0, 1, \dots, n$

$\Downarrow$

$$l_i(x_j) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Construction. Start with $l_0(x)$

$$l_0(x_0) = 1$$

$$l_0(x_j) = 0, \quad j \neq 0.$$

$$1 \leq j \leq n$$

$$l_0(x) = c(x-x_1)(x-x_2)\dots(x-x_n) =$$

$$= c \prod_{j=1}^n (x-x_j)$$

$$1 = l_0(x_0) = c \prod_{j=1}^n (x_0 - x_j)$$

$$c = \frac{1}{\prod_{j=1}^n (x_0 - x_j)}$$

$$\downarrow$$

$$l_0(x_i) = \prod_{j=1}^n \frac{x_i - x_j}{x_0 - x_j}$$

In general,

$$l_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}$$

$$i = 0, 1, 2, \dots, n.$$

Lagrange polynomial interpolation:

$$p(x) = \sum_{k=0}^n y_k l_k(x)$$

$$l_k(x) = \prod_{\substack{j=0 \\ j \neq k}}^n \frac{x - x_j}{x_k - x_j} \quad (0 \leq i \leq n).$$

Ex

x	x_0	x_1	x_2
y	y_0	y_1	y_2

$p(x) = ?$

$$p(x) = y_0 p_0(x) + y_1 p_1(x) + y_2 p_2(x)$$

$$p_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}$$

$$p_2(x) = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}$$

$$p_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}$$

Another thread :

Look for

$$p(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$$

given that $p(x_i) = y_i$, $0 \leq i \leq n$

Find: $a_0, a_1, \dots, a_n$.

$n+1$ equations

for a_i 's.

$$\underbrace{\begin{bmatrix} 1 & x_0 & x_0^2 & \dots & x_0^n \\ 1 & x_1 & x_1^2 & \dots & x_1^n \\ 1 & x_2 & x_2^2 & \dots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \dots & x_n^n \end{bmatrix}}_{\bar{W}} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} y_0 \\ y_1 \\ \vdots \\ y_n \end{bmatrix}$$

$\bar{W}$ - a Vandermonde matrix.

W is invertible by the existence theorem

Thm (Polynomial interpolation error).

Let f be a function in $C^{n+1}[a, b]$

and let p be the polynomial

of degree at most n that interpolates f at $n+1$ distinct

points $x_0, x_1, \dots, x_n$ in the interval

$[a, b]$. To each x in $[a, b]$ there

corresponds a point ξ_x in (a, b)

such that

$$(*) \quad f(x) - p(x) = \frac{1}{(n+1)!} f^{(n+1)}(\xi_x) \prod_{i=0}^n (x - x_i)$$

Pf. If $x \in \{x_0, x_1, \dots, x_n\}$ then both sides of (*) are 0. $\Rightarrow$ statement is true.

Let $x \notin \{x_0, x_1, \dots, x_n\}$, $x \in [a, b]$.

$$\text{Set } w(x) = \sum_{i=0}^n (x - x_i)$$

$$q = f - p - \lambda w, \text{ where}$$

λ is ~~the~~ real # such that

$$q(x) = 0.$$

$$\lambda = \frac{f(x) - p(x)}{w(x)}$$

Note that $q \in C^{n+1}[a, b]$ and $q = 0$

at $\underbrace{x_0, x_1, \dots, x_n}_{n+2 \text{ points}}, x$. By Rolle's thm, q' has at least $n+1$ distinct zeros on (a, b) .

Similarly, q'' has at least n distinct zeros on (a, b) .

Repeat to find that

$q^{(n)}$ has at least one zero, say,

ξ_x in (a, b) .

$$\begin{aligned} q^{(n+1)} &= f^{(n+1)} - p^{(n+1)} - \lambda w^{(n+1)} = \\ &= f^{(n+1)} - (n+1)! \lambda \end{aligned}$$

$$\deg p \leq n \Rightarrow p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$$

$$\Rightarrow p^{(n)} = n(n-1)(n-2)\dots 1 a_n$$

$$p^{(n+1)} = 0$$

$$\deg w(x) = n+1 \Rightarrow w^{(n+1)} = 1 \cdot 2 \cdot 3 \dots n(n+1) \cdot$$

leading coefficient of w .

$$w(x) = \prod_{i=0}^n (x - x_i) \Rightarrow \text{lead. coeff.} = 1.$$

Thus

$$\begin{aligned}
 0 &= \varphi^{(n+1)}(\xi_x) = f^{(n+1)}(\xi_x) - (n+1)! \lambda = \\
 &= f^{(n+1)}(\xi_x) - (n+1)! \frac{f(x) - p(x)}{w(x)}
 \end{aligned}$$

↑
(↓)

$$\frac{f^{(n+1)}(\xi_x) w(x)}{(n+1)!} = f(x) - p(x)$$

□.

[Ex]. $f(x) = \sin x$, $n=9$.

$(x_0, x_1, \dots, x_9 \in [0, 1])$

Estimate the interpolation error.

$$|f(x) - p(x)| \leq \frac{1}{10!} \left| \frac{d^{n+1}}{dx^{n+1}} \sin x \right|_{x=\xi_x}$$

$\xi_x \in (0, 1)$

$$\prod_{i=0}^9 |x - x_i| \leq \frac{1}{10!} \cdot 1$$

≤ 1 if $x \in [0, 1]$ $x_i \in [0, 1]$.

Numerical Differentiation 175

Sampling
 $f(x) \rightarrow$

$$x_0, x_1, \dots, x_n$$

$f(x_0), f(x_1), \dots, f(x_n) \sim$ snapshot of $f(x)$

$$f(x) \mapsto \bar{V} = [f(x_0), \dots, f(x_n)]$$

$$\xrightarrow{\text{or}} \bar{W} = \begin{pmatrix} x_0 & x_1 & \dots & x_n \\ f(x_0) & f(x_1) & \dots & f(x_n) \end{pmatrix}$$

Goal: create differentiation and integration formulas for ~~snapshots~~ snapshots.

Differentiation is based on Taylor formula

$$f(x+h) = f(x) + h f'(x) + \frac{h^2}{2} f''(\xi)$$

$$f'(x) = \frac{1}{h} [f(x+h) - f(x)] - \frac{h}{2} f''(\xi)$$

$\left\{ \begin{array}{l} \text{- between} \\ x \text{ and} \\ x+h \end{array} \right.$

$$f'(x) \approx \frac{1}{h} [f(x+h) - f(x)]$$

The term

$$-\frac{h}{2} f''(\xi) \text{ is}$$

the truncation

error.

$$f'(x) \stackrel{\text{def}}{=} \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Another formula:

$$f(x+h) = f(x) + h f'(x) + \frac{h^2}{2} f''(\xi_1) + \frac{h^3}{3!} f^{(3)}(\xi_1)$$

$$f(x-h) = f(x) - h f'(x) + \frac{h^2}{2} f''(x) - \frac{h^3}{3!} f^{(3)}(\xi_2)$$

// Subtract

$$f'(x) = \frac{1}{2h} [f(x+h) - f(x-h)] - \frac{h^2}{12} [f^{(3)}(\xi_1) + f^{(3)}(\xi_2)]$$

Another 1st derivative formula

$$f'(x) = \frac{1}{h^2} [f(x+h) - 2f(x) + f(x-h)] - \frac{h^2}{12} f^{(3)}(\xi)$$

We can improve on accuracy by using methods like

Richardson extrapolation.

Assume

$$f(x+h) = \sum_{k=0}^{\infty} \frac{1}{k!} h^k f^{(k)}(x)$$

$$f(x-h) = \sum_{k=0}^{\infty} \frac{1}{k!} (-1)^k h^k f^{(k)}(x)$$

$$f(x+h) - f(x-h) = 2h f'(x) + \frac{2}{3!} h^3 f^{(3)}(x) + \frac{2}{5!} h^5 f^{(5)}(x) + \dots$$

$\Downarrow$

$$f'(x) = \frac{1}{2h} [f(x+h) - f(x-h)] - \left[\frac{1}{3!} h^2 f^{(3)}(x) + \frac{1}{5!} h^4 f^{(5)}(x) + \frac{1}{7!} h^6 f^{(7)}(x) + \dots \right]$$

Consider

$$\cancel{L(x) = f(x)}$$

$$L = \varphi(h) + a_2 h^2 + a_4 h^4 + a_6 h^6 + \dots \quad (1)$$

$$L := f'(x)$$

$$\varphi(h) = \frac{1}{2h} [f(x+h) - f(x-h)]$$

Fix x . Write (this an assumption)

$$L = \varphi\left(\frac{h}{2}\right) + a_2 \frac{h^2}{4} + a_4 \frac{h^4}{16} + a_6 \frac{h^6}{64} + \dots \quad (2)$$

~~Want~~ Require (1) = (2)

Try to eliminate h^2 -term.

$$L = \varphi(h) + a_2 h^2 + \dots$$

$$4L = 4\varphi\left(\frac{h}{2}\right) + a_2 h^2 + a_4 \frac{h^4}{4} + a_6 \frac{h^6}{32} + \dots$$

subtract:

$$3L = 4\varphi\left(\frac{h}{2}\right) - \varphi(h) + 3a_4 \frac{h^4}{4} - 15a_6 \frac{h^6}{16} + \dots$$

//

$$L = \frac{4}{3} \varphi\left(\frac{h}{2}\right) - \frac{1}{3} \varphi(h) - a_4 \frac{h^4}{4} - 5a_6 \frac{h^6}{16} + \dots$$

Clearly, $L - \underbrace{\left[\frac{4}{3} \varphi\left(\frac{h}{2}\right) - \frac{1}{3} \varphi(h) \right]}_{\text{a new approximation for } f'(x)} = O(h^4)$

$$\left| \begin{aligned} \frac{4}{3} \varphi\left(\frac{h}{2}\right) - \frac{1}{3} \varphi(h) &= \frac{4}{3} \left(\frac{f\left(x+\frac{h}{2}\right) - f\left(x-\frac{h}{2}\right)}{h} \right) - \\ &\quad - \frac{1}{3} \left[\frac{f(x+h) - f(x-h)}{2h} \right]. \end{aligned} \right|$$

This process can be continued as needed.

Numerical Integration

~~178~~
179

Based on Interpolation

Approach: to compute $\int_a^b f(x) dx$ numerically,

replace $f(x)$ with $g(x)$ such that

- 1) $f(x) \approx g(x)$;
- 2) $g(x)$ is easily integrated.

$g(x)$ could be an interpolant of $f(x)$. Note also that in practice, $f(x)$ ~~is~~ ~~known~~ may be known only at certain points $x_0, x_1, \dots, x_n$. But $g(x)$ can be defined for all $x \in [a, b]$. Typically, $g(x)$ is a polynomial or a piecewise polynomial (spline) function.

For instance,

$$f(x) \approx p(x) = \sum_{i=0}^n f(x_i) l_i(x),$$

where

$$l_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} \quad (0 \leq i \leq n)$$

Recall that $p(x_k) = f(x_k)$, $k = 0, 1, \dots, n$

and

$$l_i(x_k) = \sum_{\substack{j=0 \\ j \neq i}}^n \frac{x_k - x_j}{x_i - x_j} = \begin{cases} 0 & \text{if } i \neq k \\ 1 & \text{if } i = k \end{cases}$$

Set

$$\int_a^b f(x) dx \approx \int_a^b p(x) dx = \sum_{i=0}^n f(x_i) \int_a^b l_i(x) dx$$

Thus

$$\int_a^b f(x) dx = \sum_{i=0}^n A_i f(x_i) \quad (1)$$

$$A_i = \int_a^b l_i(x) dx$$

(1) is the so-called Newton-Cotes formula (if the nodes are equally spaced).

Special case of (1) is the trapezoidal rule:

$$n=1, \quad x_0=a, \quad x_1=b$$

$$l_0 = \frac{b-x}{b-a}, \quad l_1(x) = \frac{x-a}{b-a}$$

$$A_0 = \int_a^b l_0(x) dx = \frac{1}{2}(b-a) = \int_a^b l_1(x) dx = A_1$$

$$\boxed{\int_a^b f(x) dx \approx \frac{b-a}{2} [f(a) + f(b)]} \quad (2)$$

This rule is exact for polynomials of degree at most 1.

(This follows from uniqueness of the interpolating polynomial).

Consider the error analysis (^{more} on this later).

$$\begin{aligned} f(x) - p_1(x) &= f(x) - f(a)l_0(x) - f(b)l_1(x) = \\ &= f(x) - f(a) \frac{b-x}{b-a} - f(b) \frac{x-a}{b-a} \end{aligned}$$

By Thm 2, Section 6.1, ~~the~~

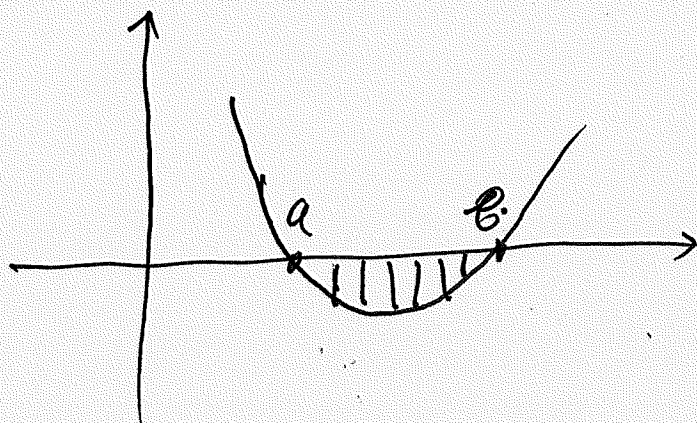
$$e(x) = f(x) - p_1(x) = \frac{1}{2!} f''(\xi_x)(x-a)(x-b)$$

Integrate:

$$\int_a^b e(x) dx = \frac{1}{2} \int_a^b f''(\xi_x)(x-a)(x-b) dx$$

Consider $S(x) = (x-a)(x-b) =$
 $= x^2 - (a+b)x + ab$

Graph:



Thus $S(x) \leq 0$ on $[a, b] \Rightarrow$

can apply mean-value theorem for integrals (look it up, if needed!)

$$\Rightarrow \int_a^b |S(x)| dx = -\frac{1}{2} f''(\xi) \int_a^b (x-a)(x-b) dx$$

for some $\xi \in [a, b]$. $\int_a^b S(x) dx =$

$$= \left. \frac{x^3}{3} - \frac{a+b}{2} x^2 + abx \right|_a^b =$$

$$= -\frac{1}{6} (b^3 + 3ab^2 - 3a^2b - a^3) = -\frac{1}{6} (b-a)^3$$

Therefore,

$$\left| \int_a^b f(x) dx - \int_a^b p(x) dx = -\frac{1}{12} (b-a)^3 f''(\xi) \right|$$

To improve accuracy, can partition $[a, b]$

as $a = x_0 < x_1 < \dots < x_n = b$ and

use the trapezoidal rule on each subinterval. This yields the

composite trapezoidal rule:

$$\int_a^b f(x) dx = \sum_{i=1}^n \int_{x_{i-1}}^{x_i} f(x) dx \approx \frac{1}{2} \sum_{i=1}^n (x_i - x_{i-1}) [f(x_{i-1}) + f(x_i)].$$

For the uniform spacing,

$$x_i = a + ih, \quad h = \frac{b-a}{n}$$

$$\int_a^b f(x) dx \approx \frac{h}{2} [f(a) + 2 \sum_{i=1}^{n-1} f(a + ih) + f(b)].$$

The error of the composite trapezoidal rule is

$$O(h^2) = -\frac{1}{12} (b-a) h^2 f''(\xi) =$$

$$\text{where } \xi \in (a, b) \quad = -\frac{1}{12} \frac{1}{h^2} (b-a)^3 f''(\xi)$$

$$\begin{aligned} h^2 &= \left(\frac{b-a}{n} \right)^2 = \\ &= \frac{(b-a)^2}{n^2} \end{aligned}$$

Method of undetermined coefficients.

Suppose that we have a formula

$$\int_a^b f(x) dx = \sum_{i=0}^n A_i f(x_i)$$

$$|A_i - ?|$$

with A_i not yet determined.

Check that

$$\int_a^b l_j(x) dx = \sum_{i=0}^n A_i l_j(x_i) = A_j$$

$$\text{since } l_j(x_i) = \delta_{ij}.$$

This allows us to derive such approximate integration formulas by requiring that they be exact for polynomials ~~that~~ of ~~the~~ degree not exceeding a certain threshold.

Ex. Suppose $f(x)$ is known at

$$x_0 = 0, \quad x_1 = \frac{1}{2} \quad \text{and} \quad x_2 = 1$$

Seek a formula

$$\int_0^1 f(x) dx \approx A_0 f(0) + A_1 f\left(\frac{1}{2}\right) + A_2 f(1)$$

By using as $f(x)$ polynomials

$1, x, \text{ and } x^2$ we find

$$\begin{cases} 1 = \int_0^1 dx = A_0 + A_1 + A_2 \\ \frac{1}{2} = \int_0^1 x dx = \frac{1}{2} A_1 + A_2 \\ \frac{1}{3} = \int_0^1 x^2 dx = \frac{1}{4} A_1 + A_2 \end{cases}$$

Solving these 3 equations for A_1, A_2, A_3 yields

$$A_0 = \frac{1}{6}, A_1 = \frac{2}{3}, A_2 = \frac{1}{6}$$

Since the resulting formula is linear, it is exact for all quadratic functions $f(x) = c_0 + c_1 x + c_2 x^2$

Simpson's rule

~~109~~
188

Calculating undetermined coefficients on $[a, b]$ with $x_0 = a$, $x_1 = \frac{a+b}{2}$, $x_2 = b$ we obtain the Simpson's rule:

$$\int_a^b f(x) dx \approx \frac{b-a}{6} \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right]$$

This is exact for polynomials of degree ≤ 2 . However, an interesting property of the Simpson's rule is that it is exact for polynomials of degree ≤ 3 .

This can be checked directly by using $f(x) = x^3$.

(Ex, (derivation of the Simpson's rule), $n=2$

$$\int_a^b f(x) dx = A_0 f(a) + A_1 f\left(\frac{a+b}{2}\right) + A_2 f(b)$$

$$1) \int_a^b 1 dx = b-a \Rightarrow \boxed{b-a = A_0 + A_1 + A_2}$$

$$2) \int_a^b x dx = \frac{x^2}{2} \Big|_a^b = \frac{b^2 - a^2}{2}$$

$$\Rightarrow \boxed{\frac{b^2 - a^2}{2} = A_0 a + A_1 \frac{a+b}{2} + A_2 b}$$

$$3) \int_a^b x^2 dx = \frac{x^3}{3} \Big|_a^b = \frac{b^3 - a^3}{3}$$

$$\boxed{\frac{b^3 - a^3}{3} = A_0 a^2 + A_1 \left(\frac{a+b}{2}\right)^2 + A_2 b^2}$$

Solve the 3 boxed equations :

$$A_0 = \frac{b-a}{6}, \quad A_1 = 4 \frac{b-a}{6}, \quad A_2 = \frac{b-a}{6}.$$

Error Analysis

190

We wish to analyze the error in the numerical integration formula

$$\int_a^b f(x) dx \approx \sum_{i=0}^n A_i f(x_i)$$

From Sect. 6.1, if $p(x)$ is an interpolating ~~polynomial~~ for $f(x)$,

$$f(x) - p(x) = \frac{1}{(n+1)!} f^{(n+1)}(\xi_x) \prod_{i=0}^n (x - x_i)$$

$$\int_a^b f(x) dx - \sum_{i=0}^n A_i f(x_i) = \frac{1}{(n+1)!} \int_a^b f^{(n+1)}(\xi_x) \prod_{i=0}^n (x - x_i) dx$$

if $|f^{(n+1)}(x)| \leq M$ on $[a, b]$, then

$$\left| \int_a^b f(x) dx - \sum_{i=0}^n A_i f(x_i) \right| \leq \frac{M}{(n+1)!} \int_a^b \prod_{i=0}^n |x - x_i| dx$$

Q: what choice of the nodes x_i minimizes the right hand side?

~~772~~
191

Fact: such node choice is

$$x_i = \frac{a+b}{2} + \frac{b-a}{2} \cos \left[\frac{(i+1)\pi}{n+2} \right] \quad i=0, 1, \dots, n$$

in $[-1, 1]$,

$$x_i = \cos \left[\frac{(i+1)\pi}{n+2} \right]$$

These x_i are zeros of

$$U_{n+1}(x) = \frac{\sin[(n+2)\theta]}{\sin \theta}, \quad x = \cos \theta$$

$\theta = \cos^{-1}(x)$

U_{n+1} is a Chebyshev polynomial of the 2^{nd} kind. We also know that

$$2^{-(n+1)} U_{n+1} = (x-x_0)(x-x_1)\dots(x-x_n)$$

$$x_i = \cos \left[\frac{(i+1)\pi}{n+2} \right]$$